

# Use Case Document: JobFlow Agent - Automated Job Application System

## Primary Use Case

### Automated Job Application Submission

Objective: Automatically apply to 320340 job postings per day across 20 students, using intelligent routing and form classification.

#### Inputs:

- Student resume, preferences, and profile data from Google Sheets
- Job listings scraped from platforms like LinkedIn, Indeed, ZipRecruiter, and Glassdoor via JobSpy

#### Process:

- Discover and classify job application forms using AI (simple vs. complex)
- Route simple forms to the HTTP Submitter
- Route complex forms to the Browser Automation Engine
- Apply with resume and cover letter generated via GPT-4
- Log application status and notify student via WhatsApp

#### Outputs:

- Logs in Google Sheets
- Daily performance report
- WhatsApp and email alerts for each application outcome

Actor: Student

## Use Case 1: Job Matching & Application

Goal: Have 1520 job applications submitted daily with minimal effort

Preconditions:

- Student profile and resume uploaded to Google Drive
- Preferences filled in Google Sheets

Triggers:

- Daily 7:00 AM CST automation run

Postconditions:

- 1520 jobs applied
- Google Sheet updated
- WhatsApp notification sent

Actor: System (JobFlow Agent)

Use Case 2: Job Discovery & Classification

Goal: Pull and evaluate 2530 jobs per student daily

Flow:

1. Load student preferences
2. Search job boards using JobSpy
3. Parse form complexity (AI Classifier)
4. Categorize forms as simple/complex

Use Case 3: Automated Job Application

Goal: Submit job applications with high success rate

Steps:

1. Use HTTP Submitter for simple forms (60% of jobs)

2. Use Browserless/Playwright for complex forms (40% of jobs)
3. Retry logic: HTTP Browser Manual Queue
4. Log and notify via Google Sheets + WhatsApp

#### Use Case 4: System Monitoring & Alerting

Actors: Admin / System Monitor

Goals:

- Ensure system health
- View live job processing
- Monitor metrics like success rate, error rate, proxy health

Tools:

- Admin Dashboard (port 8002)
- Slack Alerts, Email Alerts, Sentry, Grafana, Prometheus

#### Use Case 5: System Maintenance & Recovery

Actors: DevOps/Engineering

Goals:

- Ensure reliability
- Handle job board changes, service failures, and backups

Key Features:

- Systemd-managed services
- Daily logs and monitoring scripts
- Redis session handling
- PostgreSQL backups

#### Use Case 6: Reporting

Goal: Deliver daily and historical performance insights

### Output Includes:

- Applications submitted
- Success/failure breakdown
- Time per student
- Proxy usage and health

### Supporting Use Cases

#### AI Form Classifier:

- Determines if a job form is simple or complex
- Based on 7+ features including field count, file uploads, JS detection

#### Resume Parsing:

- Extracts candidate name, email, phone, skills, experience from PDF/DOCX resumes

#### Cover Letter Generation:

- Uses GPT-4 to auto-generate personalized letters based on job and resume content

#### Google Integration:

- Uses Google Drive API and Sheets API for student data storage and tracking

### Business Impact Use Case

- Cost per application: \$1.50
- ROI: 600900% in first year
- Payback Period: 23 months
- Staff Reduction: Removes need for manual application teams
- Scale: Easily extendable to 100+ students